Database (SOC-3020), Fall 2026



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Homework 1

Issued Monday 28 September. Due as **zipped archive** uploaded into eCLASS by **Tuesday 6 October 2026 at noon** (12:00, system's clock). Hardcopy: none (upon request only for specific cases).

Please note the following:

1. Deadline is *strict*. **Electronic deadline passed without submission implies mark 0 (zero)**. (Equivalently, you will not be admitted to enter the classroom for Quiz 1.)

2. The number of students is: 1 to 3. See next.

2.1 Teams formed by **4 members or more are not permitted** (all students are not admitted to Quiz 1).

2.2 Although teams can be formed by students from any of the sections 003 to 004, it is preferable if students in a team belong to the same section.

3. The homework must be written using LaTeX, either by installing one of following versions of it:

https://ctan.org/pkg/texlive
https://ctan.org/pkg/mactex
https://miktex.org/,

or using OVERLEAF, an easy-to-use online editor for LaTeX available at https://www.overleaf.com.

3.1 Any LaTeX template (OVERLEAF provides many) is okay. Teams are encouraged to create their own template, provided that the cover of their work contains the data required (see next).

3.2 The **cover** must contain the full name of the students in the team, their ID number and section. In particular, students shall be listed in increasing order by ID number; the name of the student who eventually uploads the homework to eCLASS shall be underlined.

4. MODALITY OF SUBMISSION:

+ One member of the team (say, the "team leader") is required to upload into eCLASS one single **zipped archive** containing:
(1) the LaTeX source file (extension .tex);
(2) its compilation (.pdf file),
(3) a folder/directory with all images or figures used in the document (if any).
+ The **name of zipped archive**, as well the **name of each submitted file** (1) to (3) must be in format: 00X-ID (Example: 001-U2310123), where:
- 00X is the section of the team's member, say S, who submits the homework to eCLASS;
- ID is the ID number of S.

5. PLAGIARISM WARNING! The homework must be your own original work. Homework that containts elements of **plagiarism or cheating**, either among teams or from existing material (e.g., from the Internet, etc.) implies a mark 0/100 **for all teammates**. (Equivalently, **all team members will not be allowed to enter the next in-class quiz**.) You may not copy source code (LaTeX, diagrams, etc.) or any other material from other people or the web. Moreover, dishonesty while discussing an academic integrity issue (i.e. lying to course staff) can result in a 'F' in the course. All violations of this collaboration policy will be referred to the appropriate University disciplinary board, with possible additional disciplinary action. Please ask the instructor (not TA!) if you are unsure.

5.1 You *are* allowed to use generative AI tools provided that you declare type and model (example: ChaptGPT, available online) at the beginning of each exercise where you have been using it.

PROBLEMS AND EXERCISES:

Exercise 1: Consider the following list of top 10 most important technology websites according to AI Mode by Google (Gemini 1.5 Pro; accessed: 2026-09-28[1]):

1. **TechCrunch**
   The global gold standard for startup news, venture capital funding, and breaking updates on Silicon Valley giants.
   https://techcrunch.com

2. **The Verge**
   Offers comprehensive coverage blending technology, science, art, and culture with detailed product reviews and multimedia content.
   https://theverge.com

3. **Wired**
   A historic publication that deeply analyzes how technology shapes culture, the economy, politics, and daily life.
   https://wired.com

4. **CNET**
   One of the oldest and most trusted websites for consumer electronics reviews, buying guides, and tech news.
   https://cnet.com

5. **Ars Technica**
   Specializes in deep technical analysis, cybersecurity, open-source software, science, and technology policy for professionals and enthusiasts.
   https://arstechnica.com

6. **Engadget**
   Focuses heavily on breaking news about gadgets, consumer electronics reviews, and personal tech gear.
   https://engadget.com

7. **Tom's Hardware**
   The premier online resource for computer hardware, PC components, performance benchmarks, and DIY building guides.
   https://tomshardware.com

8. **Gizmodo**
   A popular hub that pairs a passion for tech gadgets with design, science fiction, and investigative digital journalism.
   https://gizmodo.com

9. **VentureBeat**
   A leading source for enterprise technology, focusing heavily on artificial intelligence, machine learning, and industrial innovation.
   https://venturebeat.com

[1] Promt: "Give me the list of top 10 most important technology websites in the world (in descending order). Provide the web link for each one."

10. **Mashable**
    A global media platform covering digital culture, social media trends, entertainment, and mainstream tech news.
    https://mashable.com

Then: analyze **one of the websites listed above** (any of your choice), and "reverse engineering" the main data used (schema). If necessary to retrieve more data, navigate the page linked and explore subpages from the interface. Finally, draw an E-R diagram formalizing those data and give by yourself (in lack of data) **at least the primary key for each entity set you draw**. Other attributes for each entity set are welcome but not strictly required.

NOTE: Your diagram is required to formalize **at least four entity sets**, and **each entity set must be connected to a relationship set** (i.e., "a diamond" in the diagram).

IMPORTANT: Stick to use the notation in the illustrative E-R diagram in Figure 1. If necessary, manually tune your LaTeX source by promt-engineering an AI tool so to obtain the desired notation. Add by yourself (min, max) constraints for each relationship set you cannot reverse engineering in full (for data missing or hidden). Use common sense.

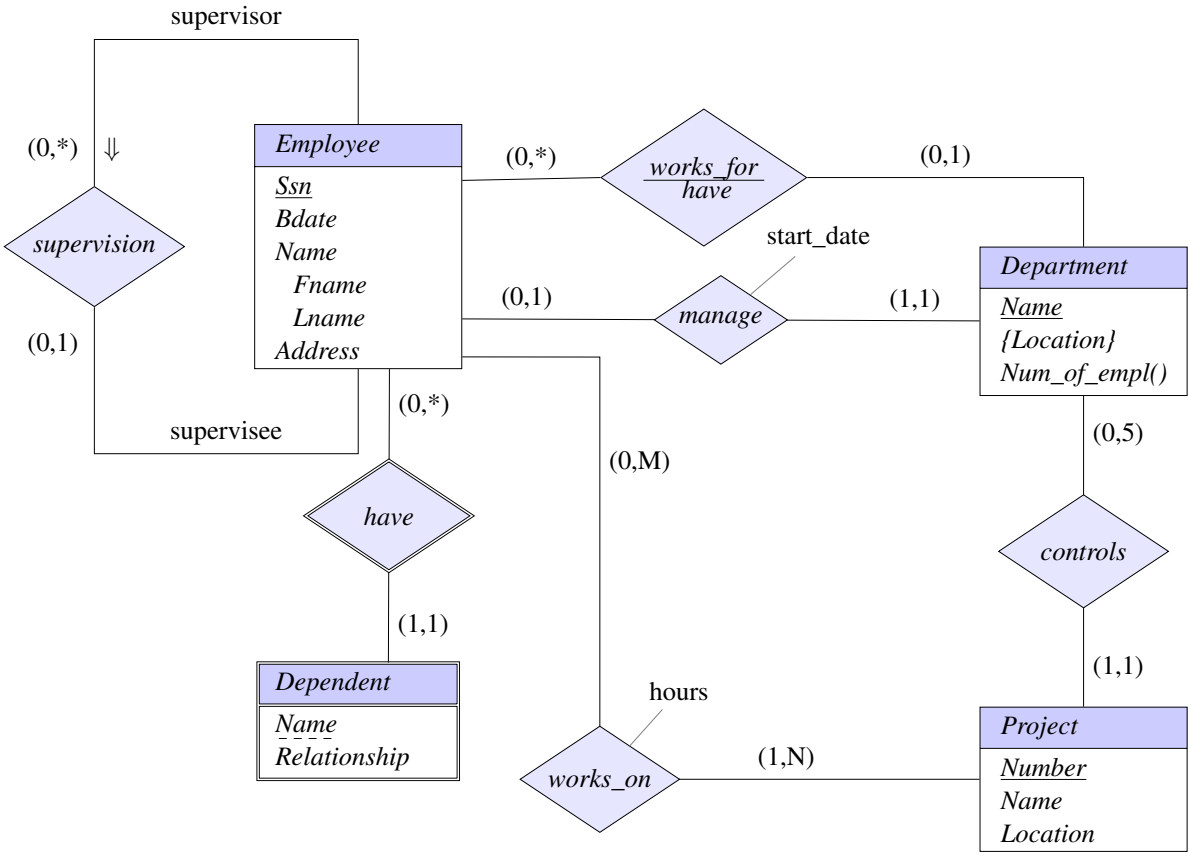


Figure 1: E-R diagram for a company DB (use it as reference for notation).

Exercise 2: Consider the following list of top 10 most important technology websites in Uzbekistan according to AI Mode by Google (Gemini 1.5 Pro; accessed: 2026-09-28[2]):

1. **IT Park Uzbekistan**
   The absolute cornerstone of the country's technology revolution, serving as the main national hub for tech startups, venture capital, and digital infrastructure.
   https://it-park.uz

2. **Kun.uz (Tech Section)**
   The dedicated technology and innovation vertical of the largest general media outlet in Uzbekistan, widely read for digital economy updates.
   https://kun.uz

3. **Gazeta.uz (Technology & Digital Segments)**
   One of the most trusted independent publications in the country, highly praised for its quality analytical articles regarding digitalization and tech policies.
   https://gazeta.uz

4. **Spot.uz**
   The premier business and commercial technology platform in Uzbekistan, serving as the go-to source for tracking fintech integrations and startup funding.
   https://spot.uz

5. **Terabayt.uz**
   Uzbekistan's top dedicated consumer technology and gadget portal, focusing heavily on hardware reviews, smartphone benchmarks, and computing news in the Uzbek language.
   https://terabayt.uz

6. **Ministry of Digital Technologies (Digital.uz)**
   The official regulatory backbone shaping the structural IT roadmap, public cloud infrastructure, and international tech partnerships across the nation.
   https://digital.uz

7. **The Tashkent Times (Tech Section)**
   The prominent English-language source for Uzbekistan's economic and innovation progress, broadcasting updates regarding international technology summits.
   https://tashkenttimes.uz

8. **Startup Uzbekistan Portal**
   The official ecosystem catalog mapping out the fastest-growing tech startups, angel investor networks, and incubator frameworks in the region.
   https://startupuzbekistan.com

9. **UzReport.news (Digital Verticals)**
   A major private media network delivering structured reporting on macroeconomic tech trends, software outsourcing, and digital banking expansion.
   https://uzreport.news

10. **Ahost.uz**
    One of the most widely utilized localized infrastructure, data center, and web domain service

[2]Promt: "Give me the list of top 10 most important technology websites in Uzbekistan (in descending order). Provide the web link for each one."

authorities handling the domestic Uzbek network segment.
https://ahost.uz

Then: analyze **one of the websites listed above** (any of your choice), and "reverse engineering" the main data used (schema). If necessary to retrieve more data, navigate the page linked and explore subpages from the interface. Finally, draw a logical schema optimized in 3NF (Third Normal Form in the relational data model) formalizing those data and give by yourself (in lack of data) **at least the primary key for each relation you draw**. Other attributes for each relation are welcome but not required.

NOTE: Your schema is required to formalize **at least four relations** ("tables"), and **each relation must be connected by a Primary Key..Foreign Key connection** to at least another relation (resulting in "an arrow" in the schema).

IMPORTANT: Stick to use the notation in the illustrative schema in Figure 2. If necessary, manually tune your LaTeX source by promt-engineering an AI tool so to obtain the desired notation. In this exercise, you can ignore (min, max) constraints. (Observe that in Figure 2 these constraints are used.)

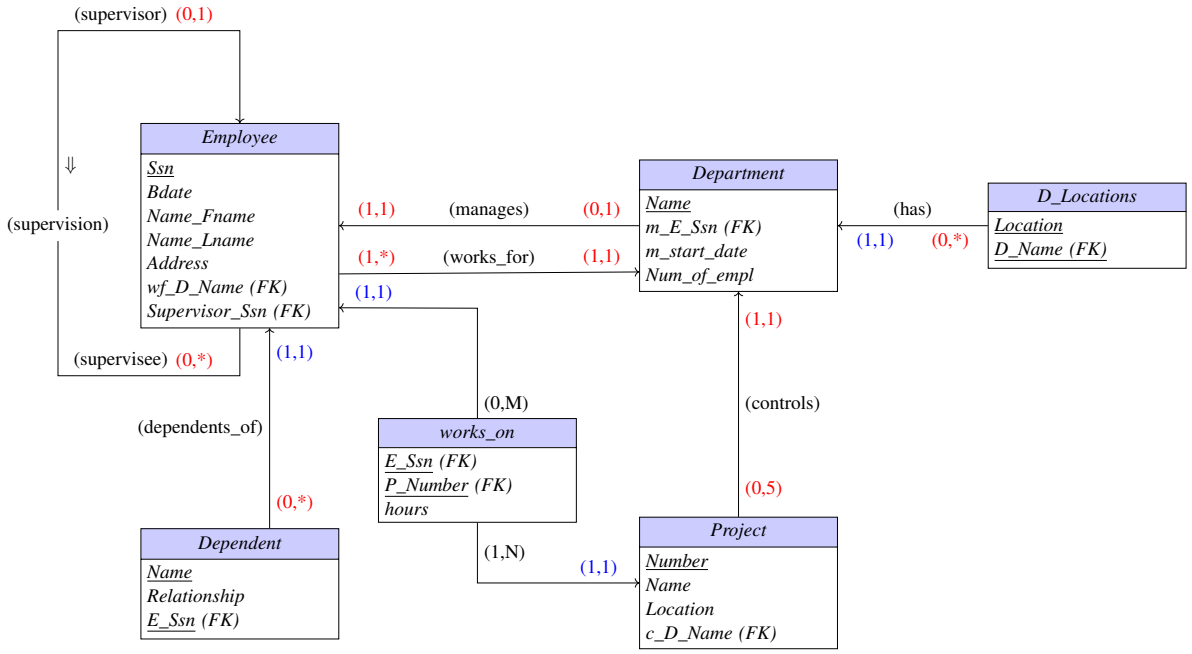


Figure 2: 3NF- Normalized Logical Schema for the company DB (use it as reference for notation).

Exercise 3: Consider the following list of top 10 most important technology websites in Europe according to ChatGPT (web version GPT-5.6 Luna; accessed: 2026-09-28[3]):

1. **Sifted** — European technology, startups, venture capital, and innovation
   https://sifted.eu/
2. **Tech.eu** — European technology, startups, funding, AI, and digital policy
   https://tech.eu/
3. **EU-Startups** — European startup news, funding, entrepreneurs, and ecosystems
   https://www.eu-startups.com/

[3]Promt: "Give me the list of top 10 most important technology websites in Europe (in descending order). Provide the web link for each one."

4. **The Next Web (TNW)** — Technology, startups, innovation, and digital culture
   https://thenextweb.com/
5. **Silicon Canals** — European technology startups, funding, and innovation
   https://siliconcanals.com/
6. **Tech Funding News** — Technology companies, venture capital, funding rounds, and startup investment
   https://techfundingnews.com/
7. **The Recursive** — Central and Eastern European technology, startups, founders, and venture capital
   https://therecursive.com/
8. **ArcticStartup** — Nordic and Baltic technology startups and growth companies
   https://arcticstartup.com/
9. **Webrazzi** — Technology, startups, venture capital, and digital business, particularly Turkey and the wider European ecosystem
   https://webrazzi.com/
10. **Trending Topics** — European technology, startups, digital business, and innovation
    https://www.trendingtopics.eu/

Then: analyze **one of the websites listed above** (any of your choice), or any other important website in academia or technology or automotive industry with domain .it (Italy), and "reverse engineering" the main data used (schema). If necessary to retrieve more data, navigate the page linked and explore sub-pages from the interface. Finally, draw a logical schema optimized in 3NF (Third Normal Form in the relational data model) formalizing those data and give by yourself (in lack of data) **at least the primary key for each relation you draw**. Other attributes for each relation are welcome but not required.

NOTE: Your schema is required to formalize **at least four relations** ("tables"), and **each relation must be connected by a Primary Key..Foreign Key connection** to at least another relation (resulting in "an arrow" in the schema).

IMPORTANT: Stick to use the notation in the illustrative schema in Figure 3 (page 7). If necessary, manually tune your LaTeX source by promt-engineering an AI tool so to obtain the desired notation. In this exercise, you can not ignore (min, max) constraints. (Observe that using the "crow's foot" notation instead of the arrows as in Figure 2 is exactly the motivation to model (min, max) constraints.) Add by yourself (min, max) constraints for each connection ("arrow" in your schema) you cannot reverse engineering in full (for data missing or hidden). Use common sense.

REMARK: This exercise is technically difficult even if you use an advanced generative AI tool. We will provide a bonus on the upcoming Quiz 1 for those students who solve this exercise satisfactorily.

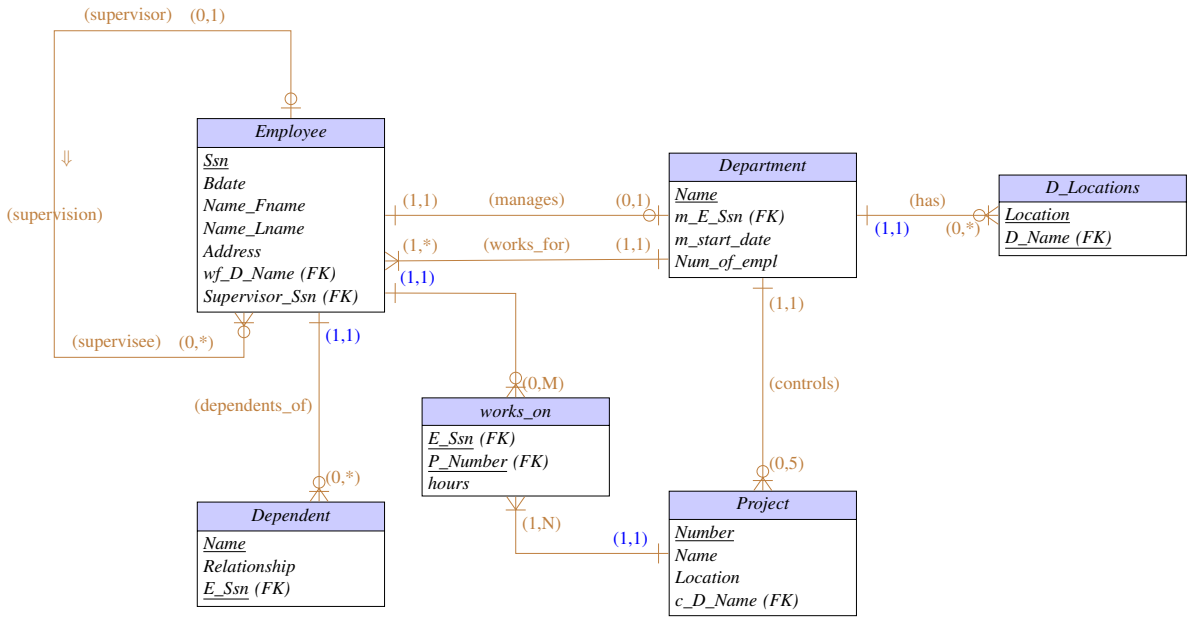


Figure 3: 3NF-Normalized Logical Schema in “crow’s foot” notation for the company DB.